

Subject:

Financial Management

BCA 5th Semester

Course Code: MIM501

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1 Introduction to Financial Management

1.1 Meaning and Definition

DEFINITION

- Financial Management is that managerial function of a business enterprise which is concerned with the planning, sourcing (procurement), deploying and controlling of the financial resources of a firm so as to achieve the organisation's financial and overall objectives.

In practice, financial management sits at the intersection of accounting and general management. It is frequently — and wrongly — treated as a synonym for accounting. Accounting is essentially a record-keeping and reporting activity: it tells a firm what has already happened to its money. Financial management, in contrast, is forward-looking and decision-oriented: it uses the data accounting produces to decide what should happen to the firm's money next. A finance manager asks questions such as 'how much capital do we need, from where should we raise it, where should it be deployed, and how do we control and monitor its use?' Financial management, therefore, concerns itself with raising both long-term and short-term capital, investing or deploying that capital efficiently, managing the resulting assets and liabilities, engaging in strategic financial planning, and continuously measuring financial performance against the plan.

1.2 Nature of Financial Management

The nature of financial management describes the essential character of the function — what kind of activity it fundamentally is. Four features capture this character.

- Analytical in Nature:** Financial management relies on quantitative tools and techniques — ratio analysis, discounted cash flow analysis, break-even analysis and budgetary control — to assess the financial performance and health of the firm before a decision is taken.
- Dynamic Process:** It is not a one-time exercise. Financial plans and decisions must continuously adapt to changes in market conditions, interest rates, regulation, competition and the internal position of the firm.
- Integrated Function:** Finance does not operate in isolation. A pricing decision from marketing, a production-capacity decision from operations, and a hiring decision from human resources all have financial consequences, so financial management must coordinate closely with every other function of the business.
- Goal-Oriented:** Every financial decision is ultimately judged against a defined financial objective — classically, the maximisation of shareholder value/wealth — rather than being taken for its own sake.

1.3 Objectives of Financial Management

The day-to-day objectives of the finance function translate the broad goal of value maximisation into concrete managerial tasks. These are distinct from the long-term 'financial goals' debate (profit vs. wealth maximisation) covered later in this module — here we are listing what a

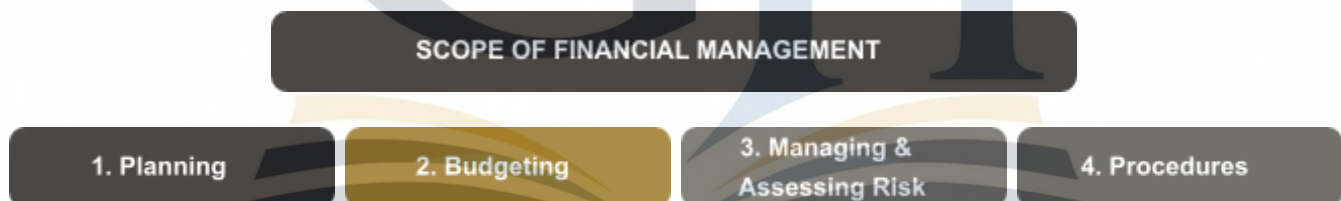
finance manager is actually trying to accomplish on an operational level.

- **Maximising Profits:** Providing insights into cost drivers — for example, flagging that a rise in raw-material prices will push up the cost of goods sold — so that pricing and procurement decisions protect the firm's margin.
- **Tracking Liquidity and Cash Flow:** Ensuring the company always has sufficient cash on hand to meet its short-term obligations as they fall due, avoiding technical insolvency even when the firm is profitable on paper.
- **Ensuring Compliance:** Keeping the firm's financial practices aligned with statutory, regulatory and industry-specific requirements.
- **Developing Financial Scenarios:** Building forecasts and 'what-if' models based on the firm's current position that account for a range of possible future market conditions.
- **Managing Relationships:** Dealing effectively and transparently with investors, lenders and the board of directors, since access to future capital depends on the trust built through this relationship.

1.4 Scope of Financial Management

The scope of financial management maps out the specific areas of activity the function actually covers. Four major areas fall within this scope.

Scope of Financial Management



- **1. Planning:** The finance manager projects how much money the company will need to maintain positive cash flow, fund growth or new products, and cope with unexpected events, sharing this information across the organisation. Planning is typically broken down into capital expenditure, travel & employee (T&E) and workforce costs, and indirect/operational expenses.
- **2. Budgeting:** The finance manager allocates the company's available funds to meet costs such as rent, salaries, raw materials and T&E, ideally leaving a margin for emergencies and new opportunities. Firms generally maintain a master budget along with sub-budgets (e.g. cash flow budget, operating budget) that may be static or flexible.
- **3. Managing and Assessing Risk:** Line managers rely on the finance function to identify and control several categories of risk: market risk (impact on investments and, for listed firms, stock performance); credit risk (customers delaying payment, hurting creditworthiness and the cost of borrowing); liquidity risk (tracking current cash flow and freeing up working capital as needed); and operational risk (a catch-all covering cyber-attacks, disaster recovery, business continuity and fraud/misconduct response).
- **4. Procedures:** The finance manager sets the written procedures governing how the finance team processes and distributes financial data (invoices, payments, reports) with security and accuracy, and defines who is authorised to take and sign off on financial decisions.

1.5 Importance of Financial Management

Sound financial management is the foundation on which three pillars of fiscal governance rest:

- **Strategising:** Identifying what needs to happen financially for the company to achieve its short- and long-term goals, using insights into current performance for scenario planning.
- **Decision-Making:** Helping business leaders choose the best way to execute plans by supplying up-to-date financial reports and data on the relevant Key Performance Indicators (KPIs).
- **Controlling:** Ensuring every department contributes to the firm's vision while operating within budget and in alignment with strategy.

REAL-WORLD BUSINESS CONNECTION

- A retail chain planning to open 20 new stores next year cannot rely on gut feeling. The finance team forecasts the capital needed (planning), allocates funds across store fit-outs and marketing (budgeting), assesses the risk of a slowdown in consumer spending (risk management), and sets clear sign-off procedures for store-level spending (procedures) — this is financial management operating across its entire scope in a single real business decision.

EXAMINATION ZONE — Introduction, Nature, Objectives & Scope

■ 1 MARK QUESTIONS

1. Define Financial Management.
2. State any two features that describe the nature of financial management.
3. Name the four major areas covered under the scope of financial management.
4. What is meant by liquidity risk?
5. Financial management is nested inside which two functions with which it is often confused?
6. Name the three pillars of fiscal governance supported by financial management.
7. What does the term "T&E" refer to in a financial planning context?

■ 5 MARK QUESTIONS

1. Explain any four objectives of financial management with suitable examples.
2. Distinguish between accounting and financial management.
3. Explain the four categories of risk a finance manager is expected to assess.
4. Why is financial management described as a "dynamic" and "integrated" function? Explain with examples.
5. Explain the role of "procedures" within the scope of financial management.

■ 15 MARK QUESTIONS

1. Define financial management and critically discuss its nature, objectives, scope and importance in the context of a modern business enterprise.
2. Financial management is often confused with accounting. Elaborate on this distinction and explain, with the help of a diagram, the four major areas that fall within the scope of financial management.

3. Discuss the various types of risks that a financial manager must plan for and assess, and explain how effective risk management contributes to the three pillars of fiscal governance.

MUST PREPARE — Highest Priority

- Define Financial Management and explain its scope with a diagram (very frequently asked).
- Difference between Accounting and Financial Management.

2 Basic Financial Decisions

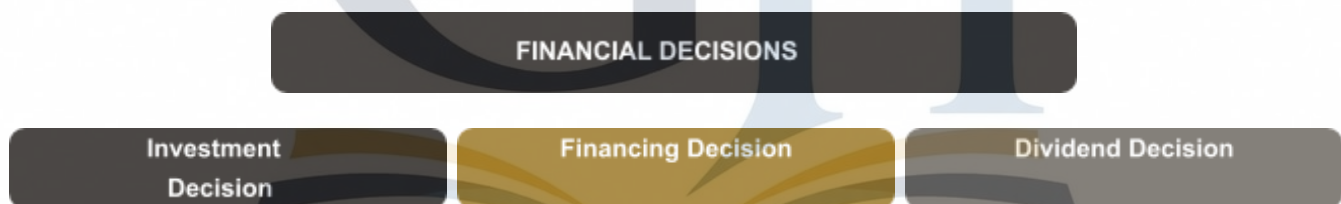
2.1 Meaning

DEFINITION

- Financial decisions are the decisions taken by managers regarding an organisation's finances — concerning the raising of money, investing it, managing assets, managing daily working capital and managing expenditure. Because these decisions directly shape the firm's assets and liabilities, the organisation's financial health is heavily dependent on the quality of these decisions.

There are three broad categories of financial decisions, and virtually every question a finance manager faces can be sorted into one of them.

Classification of Financial Decisions



2.2 Investment Decisions

Investment decisions concern how managers should invest the firm's funds across various securities, instruments and assets. These decisions determine how investment should be allocated between asset classes and which asset classes to avoid, and whether the firm should hold short-term or long-term assets, all under the organisation's own requirements and constraints. Investment decisions are generally considered the most important of the three, because they determine the firm's earning capacity and risk profile for years into the future — a poor financing decision can be refinanced, but capital sunk into the wrong asset is far harder to reverse.

2.3 Financing Decisions

Financing decisions concern how the organisation should raise the funds it needs. The general orientation of financing decisions is to raise equity while minimising reliance on debt where possible, and these decisions are typically taken in light of what the investment decision requires. Because a firm needs funds on a continuing basis, financing decisions must be revisited regularly, and the resulting capital structure should not be so rigid that it leaves no room to manoeuvre if an emergency arises or economic conditions change suddenly.

2.4 Dividend Decisions

After a firm earns a profit, it must decide how much of that profit to return to shareholders as a reward for their investment, and how much to retain within the business. Distributing too little can damage shareholder trust and confidence; distributing too much can starve the firm of funds needed for growth and reduce its margin for manoeuvre. An optimum, balanced dividend decision therefore covers both the quantum of profit to distribute and the timing of the distribution — excessive delay in paying a declared dividend damages the firm's reputation with both shareholders and the wider public.

Basis	Investment Decision	Financing Decision	Dividend Decision
Concern	Where and how to deploy funds	From where to raise funds	How much profit to distribute
Relative Importance	Considered the most critical decision	Taken in light of the investment decision	Taken after profits are earned
Key Trade-off	Risk vs. expected return of the asset	Debt (cheap, risky) vs. Equity (costly, safe)	Shareholder reward vs. retained funds for growth
Frequency	Periodic, tied to specific projects	Continuous, as funding needs arise	Usually annual / periodic

EXAMINATION ZONE — Financial Decisions

■ 1 MARK QUESTIONS

1. Name the three types of financial decisions.
2. Which financial decision is generally considered the most important, and why?
3. What does a financing decision primarily deal with?
4. What two aspects does a dividend decision cover?

■ 5 MARK QUESTIONS

1. Explain the three types of financial decisions with one example each.
2. Why are investment decisions considered more important than financing and dividend decisions?
3. Explain the factors a manager must balance while taking a dividend decision.

■ 15 MARK QUESTIONS

1. Discuss, with examples and a diagram, the three basic financial decisions taken by a firm and explain how they are interlinked.
2. Critically examine the dividend decision of a firm. What risks arise from distributing too little or too much of the firm's profits?

MUST PREPARE — Highest Priority

- Classify and explain the three financial decisions with a diagram — appears almost every year.

3 Financial Goals

3.1 Financial Goals — Overview

Financial management objectives are essential for ensuring an organisation's (or individual's) financial stability, growth and sustainability. They provide a roadmap for effectively utilising resources, managing cash flow, and achieving long-term financial goals. Core objectives include profit maximisation, cost control, maintaining liquidity, and wealth creation for stakeholders. Effective financial management also ensures timely decision-making, risk mitigation and regulatory compliance; for individuals it aids budgeting, saving and investing, while for businesses it drives operational efficiency and growth. By aligning financial decisions with strategic goals, financial management ensures optimal resource allocation and prepares the firm for unforeseen challenges. Of these goals, two are debated most intensely in theory and examinations: profit maximisation and wealth maximisation.

3.2 Profit Maximisation

DEFINITION

- Profit maximisation is the financial objective under which a firm's decisions are directed towards earning the largest possible profit, computed as Total Revenue minus Total Cost, typically over a short accounting period.

Core Profit Formulae

$$\text{Profit} = \text{TR} - \text{TC}$$

TR (Total Revenue) = Price $\times$ Quantity sold, i.e. $\text{TR} = P \times Q$

TC (Total Cost) = Fixed Cost + Variable Cost, i.e. $\text{TC} = \text{FC} + \text{VC}(Q)$ $\delta\text{TC} =$

Change in Total Cost, $\delta\text{TR} =$ Change in Total Revenue

MC (Marginal Cost) = $\delta\text{TC} / \delta Q$ MR (Marginal Revenue) = $\delta\text{TR} / \delta Q$

**SOLVED EXAMPLE 1****BASIC****Given:**

Selling price per unit (P) = Rs. 800

Total Cost function: $TC(Q) = 50,000 + 200Q$

Quantity sold (Q) = 200 units

Step 1 — Total Revenue

$$TR = P \times Q = 800 \times 200 = \text{Rs. } 1,60,000$$

Step 2 — Total Cost

$$TC = 50,000 + (200 \times 200) = 50,000 + 40,000 = \text{Rs. } 90,000$$

Step 3 — Profit

$$\text{Profit} = TR - TC = 1,60,000 - 90,000 = \text{Rs. } 70,000$$

PROFIT = Rs. 70,000**SOLVED EXAMPLE 2****CHALLENGING****Given:**

Selling price per unit (P) = Rs. 499

Fixed Cost per unit = Rs. 150

Variable Cost per unit = Rs. 100

Quantity sold (Q) = 455 units

Step 1 — Total Revenue

$$TR = P \times Q = 499 \times 455 = \text{Rs. } 2,27,045$$

Step 2 — Fixed Cost (total)

$$FC_{\text{total}} = 150 \times 455 = \text{Rs. } 68,250$$

Step 3 — Variable Cost (total)

$$VC_{\text{total}} = 100 \times 455 = \text{Rs. } 45,500$$

Step 4 — Total Cost

$$TC = FC_{\text{total}} + VC_{\text{total}} = 68,250 + 45,500 = \text{Rs. } 1,13,750$$

Step 5 — Profit

$$\text{Profit} = TR - TC = 2,27,045 - 1,13,750 = \text{Rs. } 1,13,295$$

PROFIT = Rs. 1,13,295**Solved Example 3 — Marginal Cost & Marginal Revenue
(Decision-Rule Application)**

A company's production data is given below. Calculate the Marginal Cost (MC) and Marginal Revenue (MR) for each additional unit and advise on the optimal production level.

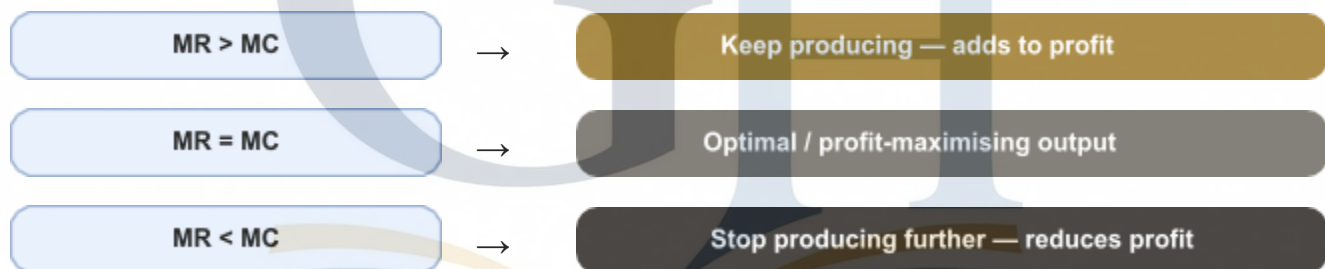
Units (Q)	Total Cost (Rs.)	Total Revenue (Rs.)
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100	5,000	7,000
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Units (Q)	Total Cost (Rs.)	Total Revenue (Rs.)
101	5,060	7,050
102	5,120	7,090
103	5,190	7,120

From Q→Q	δTC	δTR	$MC = \delta TC / \delta Q$	$MR = \delta TR / \delta Q$
100→101	60	50	Rs. 60	Rs. 50
101→102	60	40	Rs. 60	Rs. 40
102→103	70	30	Rs. 70	Rs. 30

Profit-Maximising Production Rule



Applying this rule: from unit 102 to 103, MC (Rs. 70) exceeds MR (Rs. 30), so producing the 103rd unit is not advisable — the firm's profit is not maximised beyond 102 units.

COMMON MISTAKES

- Confusing average cost/revenue with marginal cost/revenue — MC and MR must always be calculated on the change between two consecutive output levels, never on the total or average figure.
- Forgetting to compare MR with MC directly to state the decision rule — a numerical answer without the "produce / do not produce" conclusion loses marks.
- Treating Fixed Cost as if it changes with output — FC stays constant within the relevant range; only VC varies with Q.

3.3 Wealth Maximisation

DEFINITION

- Wealth maximisation is the financial objective under which a firm's decisions are directed towards maximising the market value of the shareholders' wealth — measured through the market price of the firm's shares — rather than through short-term accounting profit.



Wealth Maximisation Formula

$$\text{Wealth} = \text{MPS} \times N \mid \delta \text{Wealth} = (\text{New MPS} - \text{Old MPS}) \times N$$

MPS = Market Price per Share

N = Number of Shares Outstanding

SOLVED EXAMPLE 1

BASIC

Given:

Market Price per Share (MPS) = Rs. 150

Number of Shares Outstanding (N) = 50,000

Wealth

$$\text{Wealth} = \text{MPS} \times N = 150 \times 50,000$$

SHAREHOLDER WEALTH = Rs. 75,00,000

SOLVED EXAMPLE 2

BASIC

Given:

MPS in 2023 = Rs. 150

MPS in 2024 = Rs. 165

Number of Shares Outstanding = 50,000

Wealth in 2023

$$\text{Wealth}_{2023} = 150 \times 50,000 = \text{Rs. } 75,00,000$$

Wealth in 2024

$$\text{Wealth}_{2024} = 165 \times 50,000 = \text{Rs. } 82,50,000$$

Change in Wealth

$$\delta \text{Wealth} = 82,50,000 - 75,00,000 = \text{Rs. } 7,50,000$$

INCREASE IN WEALTH = Rs. 7,50,000 (Rs. 7.5 lakh)

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3.4 Difference Between Profit Maximisation and Wealth Maximisation

Basis	Profit Maximisation	Wealth Maximisation
Goal	Maximise a company's profit by cutting costs or raising revenue; prioritises immediate profit generation.	Maximise the wealth of all shareholders; encourages strategic investments that may not pay off immediately but will in future.
Time Horizon	Generally a short-term strategy.	Focuses on the long term.
Time Value of Money	Does not consider the time value of money.	Considers the time value of money.
Sustainability	May not always support sustainable decisions.	Uses sustainable practices.
Flexibility	Less flexible; may not adapt to short-term market shifts.	Allows easy readjustment to strategies for long-term goals.
Risk	Risky — requires high risk tolerance for immediate profit.	Comparatively less risky; seeks long-term sustainability.
ESG Considerations	Environmental, social and governance factors may not be prioritised.	ESG factors are prioritised — improves reputation, compliance and stakeholder relationships.
Measurement	Net Profit or Earnings Per Share (EPS).	Stock price and shareholders' returns.
Stakeholder Impact	Prioritises business owners / shareholders alone.	Considers shareholders, employees and society at large.
Example	A company cuts R&D spending to boost short-term profit.	A company invests in innovation for long-term growth.

3.5 Profit Maximisation vs Wealth Maximisation — Critical Analysis

Profit maximisation in financial management focuses on enhancing a company's earnings as a measure of operational efficiency. Wealth maximisation, in contrast, seeks to boost the overall value of the organisation, benefiting stakeholders over the long term. While profit maximisation addresses short-term gains, wealth maximisation is oriented toward sustainable growth, prioritising shareholder wealth and the entity's market valuation.

Both approaches carry genuine merit and neither can be dismissed outright. Without healthy profits, a company cannot sustain its operations, invest in growth, or provide returns to shareholders — profits and wealth are therefore not opposites but stages on the same path, so in a sense the two strategies go hand in hand. A company needs profits to survive, particularly in its early stages. However, focusing on profit alone can lead to poor decisions such as cutting costs at the expense of product quality or adopting unethical practices, both of which can be very harmful to the firm's reputation and long-term sustainability.

For example, a firm that switches to cheap, low-quality raw materials will increase short-term profit through lower production cost, but product quality — and with it, customer satisfaction — will decline over time. Eventually the firm's reputation suffers and it may lose a significant share of its customer base. For most companies, therefore, a balanced approach that treats profit as a

necessary stepping-stone toward wealth is the most defensible position.

The choice between the two also depends on the firm's stage and goals. A startup focused on building a market presence would likely prioritise profit maximisation to ensure it has enough cash flow to sustain daily operations and achieve quick growth. An established company aiming to be a market leader, on the other hand, is more likely to focus on wealth maximisation to increase its long-term value and enrich its shareholders.

REAL-WORLD BUSINESS CONNECTION

- A pharmaceutical company may reject a project that looks highly profitable on paper if its projected cash flows, once discounted for time and risk, produce a negative NPV — because accepting it would actually destroy shareholder wealth even though it raises accounting profit in the short run. This is the clearest real-world illustration of why wealth maximisation, not profit maximisation, drives modern investment appraisal.

EXAMINATION ZONE — Financial Goals — Profit vs Wealth Maximisation

■ 1 MARK QUESTIONS

- Define profit maximisation.
- Define wealth maximisation.
- State the formula for shareholder wealth.
- Does profit maximisation consider the time value of money?
- Which of the two goals is measured through Earnings Per Share (EPS)?
- State the marginal decision rule for profit maximisation.

■ 5 MARK QUESTIONS

- Distinguish between profit maximisation and wealth maximisation on any five parameters.
- Explain, with an example, why wealth maximisation is considered superior to profit maximisation.
- A firm sells a product at Rs. 650 per unit; $TC(Q) = 40,000 + 180Q$. Find the profit at $Q = 250$ units.
- Explain the significance of the $MR = MC$ rule in profit maximisation.

■ 15 MARK QUESTIONS

- Critically compare profit maximisation and wealth maximisation as financial goals of a firm. Support your answer with a comparative table and suitable examples.
- "Wealth maximisation is a superior goal to profit maximisation, yet a firm cannot survive without profits." Discuss this statement critically with reference to the relationship between the two goals.
- ABC Ltd. had 40,000 shares outstanding. Its market price per share was Rs. 120 in Year 1 and Rs. 138 in Year 2. Calculate the shareholder wealth in both years and the increase in wealth. Also explain how this measure differs from a profit-based measure of performance.

MUST PREPARE — Highest Priority

- Profit Maximisation vs Wealth Maximisation — comparative table (near-certain to appear).
- Critical analysis / relationship between the two goals — common 15-mark question.

4 Finance Functions & Cost of Capital

4.1 Cost of Capital

DEFINITION

- Cost of capital is the minimum rate of return, or profit, that a company must earn on an investment before it can be said to be generating value. It is calculated by a firm to determine financial risk and to judge whether a proposed investment is justified.

Company leaders use the cost of capital to gauge how much money a new venture must generate to offset its upfront cost and still deliver a profit; it is also used to analyse the potential risk of future business decisions.

Significance: Cost of capital is extremely important to investors and analysts, who use it to determine stock prices and the potential return from acquired shares. If a company's financial statements or cost of capital become volatile, the perceived cost of its shares may rise sharply, and as a result investors may become reluctant to provide financial backing.

To determine the overall cost of capital, business leaders, accounting departments and investors evaluate several individual components:

Components of Cost of Capital



All four combine, in proportion to their weight in the capital structure, into the Weighted Average Cost of Capital (WACC).

5 Net Present Value (NPV)

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5.1 Meaning

DEFINITION

- Net Present Value (NPV) is the difference between the present value of a project's cash inflows and the present value of its cash outflows over a period of time. NPV is used in capital budgeting and investment planning to analyse a project's projected profitability after adjusting for the time value of money.

Money available today is worth more than the same amount available in the future, because today's money can be invested to earn a return. NPV formally captures this by discounting every

future cash inflow back to its present-day value before comparing it against the initial investment.

NPV Formula

$$PV = FV / (1 + r)^t \quad NPV = \sum [Ct / (1 + r)^t] - C0$$

Ct = Cash flow in year "t"

r = discount rate (cost of capital / required rate of return) n = total number of years of the project

C0 = initial investment (outflow at time 0)

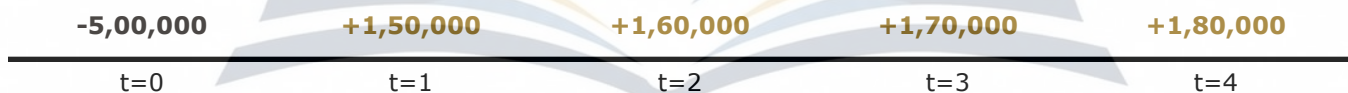
NPV Decision Rule



Solved Example 1

ABC Ltd. is planning to invest in a new machine. Initial Investment (C0) = Rs. 5,00,000; Project life = 4 years; Cost of capital = 10%. Estimated cash inflows: Year 1 = Rs. 1,50,000, Year 2 = Rs. 1,60,000, Year 3 = Rs. 1,70,000, Year 4 = Rs. 1,80,000. Calculate NPV and advise whether the company should accept the project.

Cash Flow Timeline for the Project



Discount rate r = 10%. Each future inflow is brought back to its present value before comparison with C0.

SOLVED EXAMPLE 1**BASIC****Given:**

$C_0 = \text{Rs. } 5,00,000$

Life = 4 years

$r = 10\%$

Cash inflows: Yr1=Rs. 1,50,000, Yr2=Rs. 1,60,000, Yr3=Rs. 1,70,000, Yr4=Rs. 1,80,000

Year 1

$PV_1 = 1,50,000 / (1.10)^1 = \text{Rs. } 1,36,363.64$

Year 2

$PV_2 = 1,60,000 / (1.10)^2 = \text{Rs. } 1,32,231.40$

Year 3

$PV_3 = 1,70,000 / (1.10)^3 = \text{Rs. } 1,27,805.40$

Year 4

$PV_4 = 1,80,000 / (1.10)^4 = \text{Rs. } 1,22,911.45$

Total PV of Inflows

$1,36,363.64 + 1,32,231.40 + 1,27,805.40 + 1,22,911.45 = \text{Rs. } 5,19,311.89$

NPV

$NPV = \text{Total PV of Inflows} - C_0 = 5,19,311.89 - 5,00,000 = \text{Rs. } 19,311.89$

NPV = Rs. 19,311.89 (POSITIVE) → ACCEPT THE PROJECT

Practice Problem (Challenging) — Try This Yourself

XYZ Ltd. is evaluating a project with an initial investment of Rs. 8,00,000, a life of 3 years, and a required rate of return of 12%. Expected cash inflows are Year 1 = Rs. 3,20,000, Year 2 = Rs. 3,50,000, and Year 3 = Rs. 3,80,000. Compute the NPV and recommend whether the project should be accepted. (Answer key provided in the Numerical Practice Set at the end of this module.)

COMMON MISTAKES

- Discounting cash flows using the wrong rate — always use the firm's cost of capital / required rate of return, never an arbitrary interest rate.
- Confusing cash flow with accounting profit — NPV uses actual cash inflows, not book profit (which may include non-cash items such as depreciation).
- Forgetting to subtract the initial investment C_0 after summing the present values of inflows.
- Using the wrong exponent "t" for each year — Year 1 cash flow is discounted by $(1+r)^1$, not $(1+r)^0$.

REAL-WORLD BUSINESS CONNECTION

- A company may reject a project that looks profitable in simple accounting terms — for example, one that returns more cash than it costs in total — purely because its NPV is negative once cash flows are properly discounted for time and risk. This is why NPV, not simple payback or accounting profit, is treated as the theoretically soundest investment appraisal technique.

EXAMINATION ZONE — Net Present Value (NPV)
■ 1 MARK QUESTIONS

1. Define Net Present Value.
2. State the NPV decision rule.
3. What does the symbol "C0" represent in the NPV formula?
4. What discount rate is normally used to compute NPV?
5. Is NPV based on accounting profit or cash flow?

■ 5 MARK QUESTIONS

1. Explain the concept of NPV and state its decision rule with reasons.
2. A project requires an initial investment of Rs. 3,00,000 and generates Rs. 1,20,000 per year for 3 years. At a discount rate of 10%, calculate the NPV. (Use PV factors: Yr1=0.909, Yr2=0.826, Yr3=0.751)
3. Explain why NPV incorporates the time value of money while accounting profit does not.

■ 15 MARK QUESTIONS

1. Explain the concept, formula and decision rule of Net Present Value. Illustrate with a fully solved numerical problem and a cash-flow timeline diagram.
2. Critically evaluate NPV as a technique of investment appraisal. What are its strengths and limitations compared to other techniques such as Payback Period and IRR?

MUST PREPARE — Highest Priority

- Fully solved NPV numerical with cash-flow timeline — near-guaranteed 15-mark question.
- NPV decision rule (Accept/Reject/Indifferent) — very common 1-mark question.

6 Internal Rate of Return (IRR)

6.1 Meaning

DEFINITION

- The Internal Rate of Return (IRR) is a metric used in financial analysis to estimate the profitability of potential investments. It is the discount rate that makes the Net Present Value (NPV) of all of a project's cash flows exactly equal to zero.

While NPV asks 'how much value does this project add at our given cost of capital?', IRR asks the reverse question: 'at what discount rate would this project exactly break even?' A project is worth accepting when its IRR exceeds the firm's cost of capital, because that means the project earns more than what it costs the firm to raise the funds used to finance it.

IRR Formula (Interpolation Method)

$$NPV = \sum [Cf_t / (1+r)^t] - C_0 = 0$$

$$IRR = r_1 + [NPV_1 / (NPV_1 - NPV_2)] \times (r_2 - r_1)$$

C_t = Cash flow in year "t"

r_1, r_2 = two trial discount rates (one giving positive, one giving negative NPV) NPV_1, NPV_2 =

NPV computed at r_1 and r_2 respectively

Because IRR cannot usually be solved directly through algebra, it is found by trial and error (interpolation): pick a trial rate, compute the resulting NPV; if NPV is positive, try a higher rate; if negative, try a lower rate; once one trial gives a positive NPV and another gives a negative NPV, interpolate between the two to estimate IRR.

Solved Example 1

A firm invests Rs. 80,000 in a project expected to generate cash flows of Rs. 30,000 in each of Years 1, 2 and 3. Find the IRR of the project.

Cash Flow Timeline

-80,000	+30,000	+30,000	+30,000
t=0	t=1	t=2	t=3

SOLVED EXAMPLE 1**CHALLENGING****Given:**

C_0 = Rs. 80,000

Cash inflows: Rs. 30,000 each in Years 1, 2 and 3

Trial 1 – $r = 10\%$

$$NPV = 30,000/1.10 + 30,000/1.10^2 + 30,000/1.10^3 - 80,000$$

$$= 27,272.73 + 24,793.39 + 22,539.45 - 80,000 = -Rs. 5,394.43$$

Trial 2 – $r = 5\%$

$$NPV = 30,000/1.05 + 30,000/1.05^2 + 30,000/1.05^3 - 80,000$$

$$= 28,571.43 + 27,210.89 + 25,915.13 - 80,000 = +Rs. 1,697.45$$

Interpolation

$r_1 = 5\%$, $NPV_1 = +1,697.45$ | $r_2 = 10\%$, $NPV_2 = -5,394.43$

$$IRR = 5 + [1,697.45 / (1,697.45 + 5,394.43)] \times (10 - 5)$$

$$IRR = 5 + (1,697.45 / 7,091.88) \times 5 = 5 + 1.2$$

$$IRR \approx 6.2\%$$

IRR Decision Rule

IRR > Cost of Capital



Accept the project

IRR < Cost of Capital



Reject the project

IRR = Cost of Capital



Indifferent

Basis	NPV	IRR
What it measures	Absolute rupee value added to the firm.	The break-even percentage rate of return of the project.
Decision Rule	Accept if $NPV > 0$.	Accept if $IRR > \text{cost of capital}$.
Reinvestment Assumption	Assumes cash flows are reinvested at the cost of capital.	Assumes cash flows are reinvested at the IRR itself (less realistic).
Multiple Answers	Always gives a single, unambiguous value.	May give multiple IRRs when cash flow signs change more than once.
Ranking Projects	More reliable for ranking mutually exclusive projects.	Can conflict with NPV in ranking projects of different sizes.

COMMON MISTAKES

- Stopping after one trial rate — IRR interpolation always needs two trial rates, one giving a positive and one giving a negative NPV.
- Mixing up r_1 and r_2 in the interpolation formula, or using the NPV values with the wrong sign.
- Interpreting a higher IRR alone as "better" without comparing it against the firm's cost of capital.

EXAMINATION ZONE — Internal Rate of Return (IRR)

■ 1 MARK QUESTIONS

1. Define Internal Rate of Return.
2. State the IRR decision rule.
3. What method is commonly used to calculate IRR?
4. At IRR, what is the value of NPV?

■ 5 MARK QUESTIONS

1. Explain the concept of IRR and how it is computed using the trial-and-error method.
2. Distinguish between NPV and IRR on any four parameters.
3. Why might IRR give multiple values for a single project?

■ 15 MARK QUESTIONS

1. Explain the meaning, formula and computation procedure of IRR. Solve a numerical example using the interpolation method and comment on the accept/reject decision.
2. NPV vs IRR: compare both techniques in detail, discuss situations where they may give conflicting rankings for two projects, and explain which method should be preferred and why.

MUST PREPARE — Highest Priority

- IRR by interpolation — fully solved numerical (very frequently examined).
- NPV vs IRR comparison table with conflict-in-ranking discussion.

7 Cost of Debt (Kd)

7.1 Meaning

DEFINITION

- Cost of debt (Kd) refers to the effective, after-tax interest rate a company pays on its borrowed funds — loans, debentures, credit facilities or invoice financing.

Interest paid on debt is a tax-deductible expense in most jurisdictions, so the effective cost of debt to the firm is always lower than the contractual (coupon) interest rate — this is known as the tax shield on debt. Debt may be irredeemable (perpetual, never repaid) or redeemable (repaid at a specified maturity), and the formula differs accordingly.

Cost of Debt — Irredeemable

$$\text{Irredeemable Debt: } K_d = [I \times (1-t)] / NP$$

I = Annual interest payment t =

Corporate tax rate

NP = Net Proceeds = Issue Price – Issue Expenses

Cost of Debt — Redeemable

$$\text{Redeemable Debt: } K_d = [I(1-t) + (RV-NP)/n] / [(RV+NP)/2]$$

I = Annual interest t =

Tax rate

RV = Redeemable Value NP

= Net Proceeds

n = Years to maturity

DOUBT | BELIEVE | BECOME



SOLVED EXAMPLE 1

BASIC

Given:

Debentures issued = Rs. 1,00,000 at par, 10% coupon

Tax rate = 30%

Interest

$$I = 1,00,000 \times 10\% = \text{Rs. } 10,000$$

Apply Formula

$$K_d = [10,000 \times (1 - 0.30)] / 1,00,000 = [10,000 \times 0.70] / 1,00,000 = 7,000 / 1,00,000$$

COST OF DEBT (AFTER TAX) = 7%

SOLVED EXAMPLE 2

CHALLENGING

Given:

Debentures of Rs. 1,00,000 issued at Rs. 95,000 (NP)

Coupon rate = 10%, redeemable after 5 years at face value (RV = Rs. 1,00,000)

Tax rate = 30%

Given

$$I = \text{Rs. } 10,000, t = 30\%, RV = \text{Rs. } 1,00,000, NP = \text{Rs. } 95,000, n = 5$$

Numerator

$$I(1-t) + (RV - NP)/n = 10,000(0.70) + (1,00,000 - 95,000)/5 = 7,000 + 1,000 = \text{Rs. } 8,000$$

Denominator

$$(RV + NP)/2 = (1,00,000 + 95,000)/2 = 1,95,000/2 = \text{Rs. } 97,500$$

Kd

$$K_d = 8,000 / 97,500 = 0.08205$$

COST OF DEBT = 8.21%

COMMON MISTAKES

- Forgetting the tax adjustment $(1-t)$ on interest — this is the single most common error in cost-of-debt questions.
- Using the face value instead of Net Proceeds in the denominator when the debenture is issued at a discount or premium.
- For redeemable debt, forgetting to include the amortised capital gain/loss term $(RV - NP)/n$ in the numerator.

REAL-WORLD BUSINESS CONNECTION

- Debt is usually cheaper than equity for two reasons: lenders bear lower risk than shareholders (interest must be paid before dividends, and debt holders are repaid before equity holders on liquidation), and interest is tax-deductible while dividends are not — this is precisely why the WACC formula in this module multiplies the cost of debt by $(1-T)$.

EXAMINATION ZONE — Cost of Debt

■ 1 MARK QUESTIONS

1. Define cost of debt.
2. What is meant by "tax shield" on debt?
3. State the formula for cost of irredeemable debt.

■ 5 MARK QUESTIONS

1. A company issues Rs. 2,00,000, 12% debentures at par. Tax rate is 25%. Compute the cost of debt.
2. Explain why interest is adjusted for tax while computing the cost of debt.

■ 15 MARK QUESTIONS

1. Explain the concept of cost of debt for both irredeemable and redeemable debentures. Illustrate each with a fully solved numerical example.

MUST PREPARE — Highest Priority

- Cost of Debt (redeemable) numerical — appears almost every semester.

8 Cost of Preference Capital (K_p)

8.1 Meaning

DEFINITION

- A fixed rate of dividend is payable on preference shares. Although payment of this dividend is at the discretion of the Board of Directors and there is no legal binding to pay it, this does not mean preference capital is cost-free — the cost of preference capital is a function of the dividend expected by its investors.

Unlike interest on debt, preference dividend is not tax-deductible, so — unlike K_d — the preference capital formula carries no $(1-t)$ adjustment.

Cost of Preference Capital

Irredeemable: $K_p = D_p / NP$

Redeemable: $K_p = [D_p + (RV - NP)/n] / [(RV + NP)/2]$

D_p = Annual Preference Dividend
 NP = Net Proceeds from issue
 RV = Redeemable Value
 n = Years to maturity

**SOLVED EXAMPLE 1****BASIC****Given:**

Rs. 1,00,000, 10% preference shares issued at par

Dividend

$D_p = 10\% \text{ of } 1,00,000 = \text{Rs. } 10,000$

Apply Formula

$K_p = 10,000 / 1,00,000$

COST OF PREFERENCE CAPITAL = 10%

SOLVED EXAMPLE 2**CHALLENGING****Given:**

Rs. 1,00,000, 12% preference shares issued at Rs. 95,000 Redeemable after 5 years at face value

Given

$D_p = \text{Rs. } 12,000$, $RV = \text{Rs. } 1,00,000$, $NP = \text{Rs. } 95,000$, $n=5$

Numerator

$12,000 + (1,00,000 - 95,000)/5 = 12,000 + 1,000 = \text{Rs. } 13,000$

Denominator

$(1,00,000 + 95,000)/2 = \text{Rs. } 97,500$

 K_p

$13,000 / 97,500 = 0.13333$

COST OF PREFERENCE CAPITAL = 13.33%

Basis	Cost of Debt (K_d)	Cost of Preference Capital (K_p)	Cost of Equity (K_e)
Nature of Payment	Legally binding interest	Discretionary but expected dividend	Fully discretionary dividend/growth
Tax Treatment	Tax-deductible ($1-t$ applied)	Not tax-deductible	Not tax-deductible
Risk to Investor	Lowest	Moderate	Highest
Relative Cost to Firm	Cheapest	Moderate	Most expensive

EXAMINATION ZONE — Cost of Preference Capital

■ 1 MARK QUESTIONS

1. Define cost of preference capital.
2. Is preference dividend tax-deductible?
3. State the formula for irredeemable preference capital.

■ 5 MARK QUESTIONS

1. XYZ Ltd. issues Rs. 3,00,000, 11% preference shares at Rs. 2,85,000, redeemable after 4 years at par. Compute K_p .

■ 15 MARK QUESTIONS

1. Explain the concept of cost of preference capital. Distinguish it from cost of debt and illustrate both irredeemable and redeemable cases with solved examples.

MUST PREPARE — Highest Priority

- K_p numerical (redeemable case) — commonly paired with K_d in the same question.

9 Cost of Equity (K_e)

9.1 Meaning

DEFINITION

- The cost of equity is the return that a company must realise in exchange for a given investment or project in order to satisfy its equity shareholders. When a company decides whether to take on new financing, the cost of equity determines the return it must achieve to justify the new initiative.

Because equity holders bear the residual risk of the business — they are paid only after all other claims are satisfied — the cost of equity is normally the highest of all individual costs of capital. Two methods dominate its estimation: the Capital Asset Pricing Model (CAPM) and the Gordon Growth Model (GGM).

9.2 CAPM Model

The Capital Asset Pricing Model (CAPM) describes the relationship between systematic risk (the risk that cannot be diversified away) and the expected return on an asset, particularly a stock. It establishes a linear relationship between the required return on an investment and its risk, using the expected return on the market, the risk-free rate, and the asset's sensitivity to market movements (beta).

CAPM: Relationship Between Risk and Return



Higher beta (higher systematic risk) demands a higher required return — this is the central logic of CAPM.

CAPM Formula

$$K_e = R_f + \beta(R_m - R_f)$$

R_f = Risk-free rate

β = Beta of the stock (sensitivity to market movements)

R_m = Expected market return ($R_m - R_f$) = Market risk premium

SOLVED EXAMPLE 1**BASIC****Given:**

Risk-free rate R_f = 6%

Market return R_m = 13%

Beta of stock β = 1.2

Apply Formula

$$K_e = 6\% + 1.2 \times (13\% - 6\%) = 6\% + 1.2 \times 7\% = 6\% + 8.4\%$$

COST OF EQUITY (CAPM) = 14.4%

Limitations of CAPM: it rests on several unrealistic assumptions (frictionless markets, all investors holding diversified portfolios, a single-period horizon) and relies on a purely linear interpretation of the risk-return relationship. Despite these issues, CAPM remains widely used because it is simple to apply and allows easy comparison across investment alternatives.

9.3 Gordon Growth Model (GGM)

The Gordon Growth Model is a tool used to determine the intrinsic value of a stock (and, when rearranged, the cost of equity) based on a series of dividends that grow at a constant rate indefinitely. It is best applied to companies with stable, predictable dividend growth, since it assumes the growth rate continues forever.

Gordon Growth Model Formula

$$K_e = (D_1 / P_0) + g \quad \text{where } D_1 = D_0(1+g)$$

D_1 = Expected dividend next year D_0 =

Dividend just paid

P_0 = Current market price per share g =

Constant dividend growth rate

SOLVED EXAMPLE 2**BASIC****Given:**

Last dividend paid $D_0 = \text{Rs. } 5$

Growth rate $g = 6\% \text{ p.a.}$

Current market price $P_0 = \text{Rs. } 50$

Step 1 — Next Dividend

$$D_1 = D_0(1+g) = 5 \times 1.06 = \text{Rs. } 5.30$$

Step 2 — Apply Formula

$$K_e = (5.30/50) + 0.06 = 0.106 + 0.06 = 0.166$$

COST OF EQUITY (GGM) = 16.6%

Limitations of GGM: it relies heavily on the assumption of **stable**, constant dividend growth, and breaks down mathematically if the required rate of return is lower than the assumed growth rate. As a valuation tool, if the model's calculated intrinsic value exceeds the current market price, the stock is considered undervalued (a signal to buy); if it is lower, the stock is considered overvalued (a signal to sell).

Basis	CAPM	Gordon Growth Model
Core Input	Beta, risk-free rate, market return	Dividend, growth rate, market price
Best Suited For	Listed stocks with observable beta / market data	Firms with stable, predictable dividend growth
Considers Risk Explicitly?	Yes — via beta (systematic risk)	No explicit risk term — risk is embedded in the discount rate implicitly
Key Limitation	Rests on unrealistic market assumptions	Breaks down if $g \geq$ required return

COMMON MISTAKES

- Using D_0 (the dividend already paid) instead of D_1 (the expected next dividend) in the Gordon Growth formula.
- In CAPM, subtracting in the wrong order — the market risk premium is $(R_m - R_f)$, not $(R_f - R_m)$.
- Forgetting to convert the growth rate "g" to decimal form before adding it to D_1/P_0 .

EXAMINATION ZONE — Cost of Equity — CAPM & Gordon Growth Model**■ 1 MARK QUESTIONS**

1. State the CAPM formula.
2. State the Gordon Growth Model formula for cost of equity.
3. What does beta measure in the CAPM model?
4. What is meant by the market risk premium?

■ 5 MARK QUESTIONS

1. A stock has $R_f=5\%$, $R_m=12\%$, $\beta=1.5$. Compute the cost of equity using CAPM.
2. A company just paid a dividend of Rs. 8 per share, expected to grow at 5% annually; current market price is Rs. 80. Find the cost of equity using GGM.
3. Explain the meaning and limitations of the CAPM model.

■ 15 MARK QUESTIONS

1. Explain CAPM and the Gordon Growth Model as methods of estimating the cost of equity. Compare their assumptions and limitations, and illustrate each with a fully solved numerical example.
2. Critically discuss why cost of equity is usually the highest component of a firm's cost of capital, linking your answer to the risk-return relationship underlying CAPM.

MUST PREPARE — Highest Priority

- CAPM numerical — extremely high-frequency question.
- K_e + CAPM integrated question (combine formula derivation with a solved numerical).

10 Cost of Retained Earnings (K_r)

10.1 Meaning

DEFINITION

- The cost of retained earnings is the return prevailing in the market for common stock of equal risk to that already outstanding. It is also called the cost of internal capital or the cost of internal equity, because retained earnings represent profit the firm keeps rather than distributing to shareholders.

The cost of retained earnings can be computed using one of three methods — the Capital Asset Pricing Model, the dividend-yield-plus-growth-rate (discounted cash flow) method, or the bond-yield-plus-risk-premium method. Since these methods normally produce different results, an average of the three is often used for the weighted average cost of capital.

Retained earnings are not literally "free" simply because the firm does not pay a market price to raise them. There is an opportunity cost: the profit retained could instead have been distributed to shareholders, who could have reinvested it elsewhere at the going rate of return (K_e). Shareholders pay personal tax on the dividend they would have received, so the effective cost of retained earnings is slightly lower than the cost of equity, adjusted for the investor's personal tax

Cost of Retained Earnings Formula

$$K_r = K_e \times (1 - t)$$

K_e = Cost of Equity

t = Tax rate applicable to investors (personal tax on dividend income)

rate.



SOLVED EXAMPLE 1

BASIC

Given:

Cost of Equity $K_e = 15\%$

Tax rate applicable to shareholders = 30%

Apply Formula

$$K_r = 15\% \times (1 - 0.30) = 15\% \times 0.70$$

COST OF RETAINED EARNINGS = 10.5%

REAL-WORLD BUSINESS CONNECTION

- A profitable company decides to retain Rs. 50 lakh instead of paying it out as a dividend, in order to self-fund an expansion. Even though no cash actually leaves the business to 'pay' for these funds, the firm still incurs an opportunity cost equal to what shareholders could have earned had that money been distributed and reinvested elsewhere — this is exactly why K_r must still be included when computing WACC.

EXAMINATION ZONE — Cost of Retained Earnings

■ 1 MARK QUESTIONS

1. Define cost of retained earnings.
2. Why is cost of retained earnings also called the cost of internal equity?
3. State the formula relating K_r to K_e .

■ 5 MARK QUESTIONS

1. If $K_e = 18\%$ and the applicable investor tax rate is 25%, compute K_r .
2. Explain why retained earnings are not a cost-free source of capital despite involving no external issue.

■ 15 MARK QUESTIONS

1. Explain the concept and computation of cost of retained earnings, and discuss the three methods used to estimate it. Why does an opportunity cost arise even though no funds are externally raised?

MUST PREPARE — Highest Priority

- K_r numerical (simple, always paired with K_e) — near-certain to appear.

11 Weighted Average Cost of Capital (WACC)

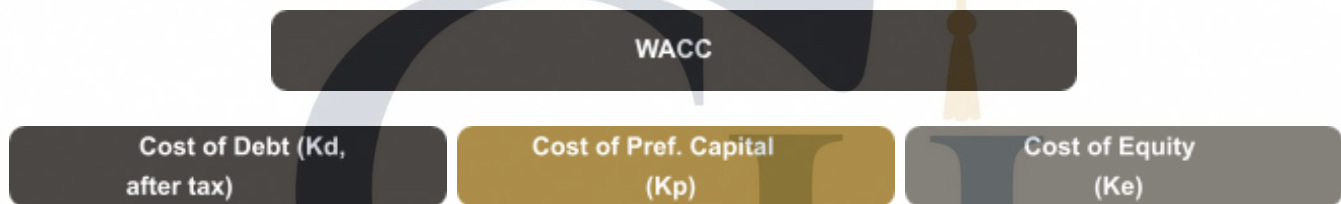
11.1 Meaning

DEFINITION

- Weighted Average Cost of Capital (WACC) represents a company's overall cost of capital, with each category of capital — debt, preference and equity — proportionately weighted by its share of the total capital structure.

WACC is calculated by multiplying the cost of each capital source by its relevant weight in terms of market value, and adding the results together. WACC is commonly used as a hurdle rate against which companies and investors gauge the desirability of a given project or acquisition, and it is the discount rate most often used for future cash flows in discounted cash flow (NPV) analysis — closing the loop back to Chapter 5 of this module.

How WACC Is Built



Each cost is weighted by its share of Total Capital ($V = E + D + P$) before being summed.

WACC Formula

$$\text{WACC} = (E/V \times K_e) + (D/V \times K_d \times (1-T)) + (P/V \times K_p)$$

E = Market value of Equity D =

Market value of Debt

P = Market value of Preference Capital V =

Total Capital = $E + D + P$

K_e, K_d, K_p = Cost of Equity, Debt, Preference Capital T =

Corporate Tax Rate

GREAT HEIGHTS

BUILD | BELIEVE | BECOME

SOLVED EXAMPLE 1**BASIC****Given:**

Equity (E) = Rs. 6,00,000 → $K_e = 12\%$

Debt (D) = Rs. 3,00,000 → $K_d = 10\%$

Preference (P) = Rs. 1,00,000 → $K_p = 9\%$

Total Capital

$V = 6,00,000 + 3,00,000 + 1,00,000 = \text{Rs. } 10,00,000$

Weighted Cost — Equity

$(6,00,000/10,00,000) \times 12\% = 0.6 \times 12\% = 7.2\%$

Weighted Cost — Debt

$(3,00,000/10,00,000) \times 10\% = 0.3 \times 10\% = 3.0\%$

Weighted Cost — Preference

$(1,00,000/10,00,000) \times 9\% = 0.1 \times 9\% = 0.9\%$

WACC

$\text{WACC} = 7.2\% + 3.0\% + 0.9\%$

WACC = 11.1%

Practice Problem (Challenging) — Try This Yourself

A company's capital structure consists of Equity Rs. 12,00,000 ($K_e=14\%$), Debt Rs. 6,00,000 (pre-tax $K_d=11\%$, tax rate=30%), and Preference Capital Rs. 2,00,000 ($K_p=10\%$). Compute the WACC, remembering to apply the tax adjustment to the cost of debt. (Answer key provided in the Numerical Practice Set at the end of this module.)

COMMON MISTAKES

- Forgetting to apply $(1-T)$ to K_d within the WACC formula, even though it may already have been applied once while computing K_d itself in Chapter 7 (use the pre-tax K_d here, or apply the tax adjustment exactly once — never twice).
- Using book values of E, D and P instead of market values when market values are available or implied.
- Using incorrect weights that do not sum to 1 (i.e. forgetting to divide each component by $V = E+D+P$).
- Adding the individual costs directly without weighting them by their proportion in the capital structure.

REAL-WORLD BUSINESS CONNECTION

- A company evaluating a new plant will only proceed if the plant's expected IRR exceeds its WACC — WACC functions as the minimum acceptable return, or hurdle rate, below which a project destroys shareholder wealth even if it appears profitable in isolation. This is the same reasoning applied by private equity firms and corporate boards worldwide when screening acquisition targets.

EXAMINATION ZONE — Weighted Average Cost of Capital (WACC)

■ 1 MARK QUESTIONS

1. Define WACC.
2. State the WACC formula.
3. Why is $(1-T)$ applied to K_d in the WACC formula?
4. What is meant by WACC acting as a "hurdle rate"?

■ 5 MARK QUESTIONS

1. A firm has Equity Rs. 4,00,000 ($K_e=15\%$) and Debt Rs. 2,00,000 (pre-tax $K_d=12\%$, tax=25%). Compute WACC.
2. Explain why market values, rather than book values, are preferred while computing WACC weights.

■ 15 MARK QUESTIONS

1. Explain the concept, formula and significance of WACC. Illustrate its computation with a fully solved numerical example covering equity, debt and preference capital.
2. Cost of Capital + WACC: Explain each of K_d , K_p , K_e , K_r individually and show, with a diagram and a numerical illustration, how they are combined into the firm's WACC.
3. Financing Decision + Cost of Capital: Discuss how a firm's choice between debt and equity financing affects its overall WACC, and explain why WACC is used as the discount rate for evaluating new projects.

MUST PREPARE — Highest Priority

- WACC numerical (equity + debt + preference) — the single most important numerical in this module.
- WACC as hurdle rate / link to NPV — very common integrated 15-mark question.

12 Numerical Masterclass

This section consolidates every numerical technique in the module into a single ready-reference sequence: Formula → Procedure → Worked Answer Reference → Practice Problem. Use it in the final days before an examination to drill calculation speed rather than re-reading full explanations.

Profit

Formula: Profit = TR – TC, where TR = $P \times Q$, TC = FC + VC(Q).

Procedure: 1) Compute TR. 2) Compute TC. 3) Subtract.

Worked Example: See Ch.3, Examples 1 & 2.

Practice Problem: A firm sells at Rs. 350/unit; FC=Rs. 20,000, VC=Rs. 150/unit; $Q=300$. Find profit.

Marginal Cost / Revenue

Formula: MC = $\partial TC / \partial Q$, MR = $\partial TR / \partial Q$.

Procedure: 1) Take consecutive Q values. 2) Find change in TC and TR. 3) Divide by change in Q. 4) Compare MR vs MC.

Worked Example: See Ch.3, Example 3.

Practice Problem: Using $Q=200 \rightarrow 201$, TC moves Rs. 9,000 $\rightarrow$ Rs. 9,055, TR moves Rs. 12,000 $\rightarrow$ Rs. 12,040. Find MC and MR.

Wealth

Formula: $\text{Wealth} = \text{MPS} \times N$; $\delta \text{Wealth} = (\text{New MPS} - \text{Old MPS}) \times N$.

Procedure: 1) Multiply MPS by number of shares for each period. 2) Subtract to find change.

Worked Example: See Ch.3, Examples 1 & 2.

Practice Problem: MPS rises from Rs. 90 to Rs. 104 for a firm with 25,000 shares. Find the increase in wealth.

NPV

Formula: $\text{NPV} = \sum [C_t / (1+r)^t] - C_0$.

Procedure: 1) Discount each year's cash flow to PV. 2) Sum all PVs. 3) Subtract C_0 . 4) Apply decision rule.

Worked Example: See Ch.5, Example 1.

Practice Problem: $C_0 = \text{Rs. } 8,00,000$; $r=12\%$; inflows Rs. 3,20,000/Rs. 3,50,000/Rs. 3,80,000 (Yr1-3). Find NPV.

IRR

Formula: $\text{IRR} = r_1 + [\text{NPV}_1 / (\text{NPV}_1 - \text{NPV}_2)] \times (r_2 - r_1)$.

Procedure: 1) Try a rate; compute NPV. 2) Try a second rate on the opposite sign. 3) Interpolate.

Worked Example: See Ch.6, Example 1.

Practice Problem: $C_0 = \text{Rs. } 50,000$; inflow Rs. 20,000 each year for 3 years. Estimate IRR (try $r=10\%$ and $r=15\%$).

Cost of Debt (Kd)

Formula: Irredeemable: $I(1-t)/NP$. Redeemable: $[I(1-t) + (RV - NP)/n] / [(RV + NP)/2]$.

Procedure: 1) Find annual interest I. 2) Apply tax adjustment. 3) Use NP (and RV, n if redeemable).

Worked Example: See Ch.7, Examples 1 & 2.

Practice Problem: Rs. 5,00,000, 9% debentures issued at Rs. 4,80,000, redeemable after 4 years at par, $\text{tax}=25\%$. Find K_d .

Cost of Preference (Kp)

Formula: Irredeemable: D_p/NP . Redeemable: $[D_p + (RV - NP)/n] / [(RV + NP)/2]$.

Procedure: 1) Find annual preference dividend. 2) Use NP (and RV, n if redeemable). No tax adjustment.

Worked Example: See Ch.8, Examples 1 & 2.

Practice Problem: Rs. 2,00,000, 11% preference shares issued at Rs. 1,90,000, redeemable after 6 years at par. Find K_p .

CAPM (Ke)

Formula: $K_e = R_f + \beta(R_m - R_f)$.

Procedure: 1) Find market risk premium ($R_m - R_f$). 2) Multiply by beta. 3) Add R_f .



Worked Example: See Ch.9, Example 1.

Practice Problem: $R_f=7\%$, $R_m=14\%$, $\beta=0.8$. Find K_e .

Gordon Growth (K_e)

Formula: $K_e = (D_1/P_0) + g$, $D_1=D_0(1+g)$.

Procedure: 1) Grow D_0 by $(1+g)$ to get D_1 . 2) Divide by P_0 . 3) Add

g . Worked Example: See Ch.9, Example 2.

Practice Problem: $D_0=Rs. 4$, $g=7\%$, $P_0=Rs. 40$. Find K_e .

Retained Earnings (K_r)

Formula: $K_r = K_e \times (1-t)$.

Procedure: 1) Take K_e . 2) Multiply by $(1 \text{ minus investor tax rate})$.

Worked Example: See Ch.10, Example 1.

Practice Problem: $K_e=20\%$, investor tax rate=20%. Find K_r .

WACC

Formula: $WACC = (E/V)K_e + (D/V)K_d(1-T) + (P/V)K_p$.

Procedure: 1) Find $V=E+D+P$. 2) Weight each cost by its share of V . 3) Sum.

Worked Example: See Ch.11, Example 1.

Practice Problem: $E=Rs. 12,00,000$ ($K_e=14\%$), $D=Rs. 6,00,000$ (pre-tax $K_d=11\%$, tax=30%), $P=Rs. 2,00,000$ ($K_p=10\%$). Find WACC.

GREAT HEIGHTS

BUILD | BELIEVE | BECOME

12.1 Answer Key — Practice Problems Referenced Above

Topic	Practice Problem	Final Answer
Profit	Rs. 350/unit; FC=Rs. 20,000; VC=Rs. 150/unit; Q=300	Profit = Rs. 40,000
MC/MR	Q=200→201 (TC Rs. 9,000→Rs. 9,055; TR Rs. 12,000→Rs. 12,040)	MC=Rs. 55, MR=Rs. 40 (MC>MR: do not produce the 201st unit)
Wealth	MPS Rs. 90→Rs. 104, N=25,000	Increase in Wealth = Rs. 3,50,000
NPV	C0=Rs. 8,00,000; r=12%; inflows 3,20,000/3,50,000/3,80,000	NPV ≈ -Rs. 7,614 (NEGATIVE → REJECT)
IRR	C0=Rs. 50,000; Rs. 20,000/yr for 3 yrs	IRR ≈ 9.7% (between trial rates 10% & 15%; refine with closer trials)
Kd	Rs. 5,00,000 @9%, NP=Rs. 4,80,000, RV=Rs. 5,00,000, n=4, t=25%	Kd ≈ 7.55%
Kp	Rs. 2,00,000 @11%, NP=Rs. 1,90,000, RV=Rs. 2,00,000, n=6	Kp ≈ 11.37%
CAPM	Rf=7%, Rm=14%, β=0.8	Ke = 12.6%
GGM	D0=Rs. 4, g=7%, P0=Rs. 40	Ke = 17.7%
Kr	Ke=20%, t=20%	Kr = 16%
WACC	E=12L(14%), D=6L(pre-tax 11%,t=30%), P=2L(10%)	WACC = 11.685%

Work through every practice problem yourself before checking this table — the discipline of setting up the formula and substituting values correctly under time pressure is the actual examination skill being tested.

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Formula Memory System

Group A — Revenue / Cost / Profit

- $TR = P \times Q$
- $TC = FC + VC(Q)$
- Profit = TR – TC
- $MC = \delta TC / \delta Q$, $MR = \delta TR / \delta Q$

Memory Cue: "Total minus Total = Profit; change over change = Marginal."

Group B — Wealth

- Wealth = MPS × N
- $\delta \text{Wealth} = (\text{New MPS} - \text{Old MPS}) \times N$

Memory Cue: "Wealth is Price times Pieces (shares)."

Group C — Present Value / NPV

- $PV = FV/(1+r)^t$
- $NPV = \sum PV \text{ of inflows} - C_0$

Memory Cue: "NPV = what it's worth today, minus what you paid today."

Group D — IRR

- $IRR = r_1 + [NPV_1/(NPV_1 - NPV_2)] \times (r_2 - r_1)$

Memory Cue: "Low rate gives positive NPV, high rate gives negative NPV — interpolate between them."

Group E — Cost of Debt

- Irredeemable: $I(1-t)/NP$
- Redeemable: $[I(1-t) + (RV - NP)/n] / [(RV + NP)/2]$

Memory Cue: "Debt always gets the tax shield — multiply interest by $(1-t)$."

Group F — Preference Capital

- Irredeemable: D_p/NP
- Redeemable: $[D_p + (RV - NP)/n] / [(RV + NP)/2]$

Memory Cue: "Same shape as debt, but NO tax adjustment — preference dividend isn't tax-deductible."

Group G — CAPM

- $K_e = R_f + \beta(R_m - R_f)$

Memory Cue: "Risk-free plus Beta times the extra the market pays for risk."

Group H — Gordon Growth

- $K_e = (D_1/P_0) + g$
- $D_1 = D_0(1+g)$

Memory Cue: "Dividend yield PLUS growth."

Group I — Retained Earnings

- $K_r = K_e \times (1-t)$

Memory Cue: "Retained earnings = Equity cost, shrunk by the investor's personal tax."

Group J — WACC

- $WACC = (E/V)K_e + (D/V)K_d(1-T) + (P/V)K_p$

Memory Cue: "Weight each cost by its slice of the capital pie, then add — only debt gets the tax shield inside WACC."